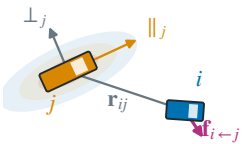
(a) Ordered source-body field



$c_i\downarrow$, $c_j\rightarrow$	P	B	C	T
P	θ	θ	θ	θ
B	θ	θ	θ	θ
C	θ	θ	θ	θ
T	θ	θ	θ	θ

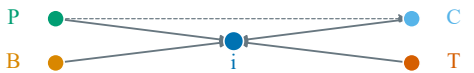
$$D_{ij}=\mathrm{diag}(\sigma_{\parallel}^{-2},\sigma_{\perp}^{-2})$$
$$\mathbf{r}_{ij}=\mathbf{p}_i-\mathbf{p}_j,\quad \mathbf{z}_{ij}=R_j^{\top}\mathbf{r}_{ij}$$
$$\bar{s}_{ij}=[-\mathbf{r}_{ij}^{\top}\Delta\mathbf{v}_{ij}/\max(\|\mathbf{r}_{ij}\|,\epsilon)]_+$$
$$U_{ij}=A_{c_ic_j}(1+\kappa_{c_ic_j}\bar{s}_{ij})e^{-\mathbf{z}_{ij}^{\top}D_{ij}\mathbf{z}_{ij}/2}$$
$$\mathbf{f}_{i\leftarrow j}=-\nabla_{\mathbf{p}_i}U_{ij}|_{\bar{s},R_j}=U_{ij}R_jD_{ij}\mathbf{z}_{ij}$$

$A,\kappa=\mathrm{softplus}(a,k),\quad \sigma=\max(0.2,e^{w})$

Same fitted θ

(b) Force-conditioned graph

$$h_i^0=\mathrm{ReLU}(W_0[\mathrm{GRU}(X_i);E_{c_i}]+b_0)$$



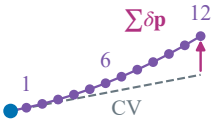
8 observed samples; top-8 incoming edges within 30 m

$$\mathbf{g}_{ji}=[\mathbf{z}_{ij}/30,\|\mathbf{r}_{ij}\|/30,R_j^{\top}\Delta\mathbf{v}_{ij}]$$
$$x_{ji}=[\mathbf{g}_{ji},R_j^{\top}\mathbf{f}_{i\leftarrow j},E_{c_ic_j}]$$
$$q_{ji}^{\ell}=M_{\ell}([h_j^{\ell},h_i^{\ell},x_{ji}])$$
$$\alpha_{ji}^{\ell}=\mathrm{softmax}_{j\rightarrow i}\,a_{\ell}(q_{ji}^{\ell})$$
$$h_i^{\ell+1}=\mathrm{GRU}_{\ell}(\sum_{j\rightarrow i}\alpha_{ji}^{\ell}q_{ji}^{\ell},h_i^{\ell})$$

Two message layers; attention $\alpha\neq\|\mathbf{f}\|$

(c) Point forecast

MLP: 64 to 24



$$\hat{\mathbf{p}}_i^k=\mathbf{p}_i+k\Delta t\,\mathbf{v}_i+\sum_{\ell=1}^{k}\delta\mathbf{p}_i^{\ell}$$

$k=1:12,\ \Delta t=0.4\,\mathrm{s}$

4.8 s; positions, not yaw

Predicted centres

(d) Current and future field synthesis



$$C(\mathbf{F})=\min(0,\mathbf{t}_r^{\top}\Pi_{b_{\max}}(g\mathbf{F}))\,\mathbf{t}_r$$
$$\mathbf{b}_S=C(\sum_j\mathbf{f}_{i\leftarrow j})$$
$$\mathbf{b}_P=\Pi_2[\sum_{k=1}^{8}\omega_kC(\sum_j\hat{\mathbf{f}}_{i\leftarrow j}^{k})]$$
$$\omega_k=0.7^k/\sum_{\ell=1}^{8}0.7^{\ell};\ \text{control horizon 3.2 s}$$

Future query: receiver position/velocity held at the current state

(e) Route-constrained execution

$$v_{\mathrm{tar}}=[v_{\mathrm{geo}}+\min(0,\mathbf{t}_r^{\top}\tilde{\mathbf{b}}_S)+\min(0,\mathbf{t}_r^{\top}\tilde{\mathbf{b}}_P)]_{[0,v_r]}$$

Commit damping and norm caps in $\tilde{\mathbf{b}}$

Go/creep + latch → Reference shaper

Safety caps bypass comfort

Soft CBF-QP

$$\min_{\mathbf{u},\xi\geq0}\|\mathbf{u}-\mathbf{u}_{\mathrm{ref}}\|^2+\rho\|\xi\|^2$$
$$A_m\mathbf{u}\leq b_m+\xi\ \ \text{(moving)}$$
$$H\mathbf{u}\leq b_h\ \ \text{(geometry/speed/no-reverse)}$$

Stock Ackermann tracker

Assigned route; no semantic lane changes

20 Hz state/control; 2.5 Hz prediction refresh. Delivered safety tracks remain independent of graph eligibility.
P: pedestrian B: bicycle C: car T: truck. Schematic mechanism; field force is not attention or probability.